

Three Solutions for the Compensation Problem*

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A model of fair division in which differential claims are grounded in talents and handicaps is studied, and a characterization of three solutions is provided. The first two solutions, conditional equality and egalitarian-equivalence, display dual properties. The consistency axiom is shown to have strong consequences in one-good models. *Journal of Economic Literature* Classification Number(s): D63, D71, I31. © 1995 Academic Press, Inc.

1. INTRODUCTION

The problem of differential claims has been dealt with in the fair division literature in a very specific way. The claims have been supposed to be expressed in the same unit as the resource to be distributed among agents (see, e.g., O'Neill [7], Aumann and Maschler [2], and Chun [3]). This is appropriate for describing problems of bankruptcy, when creditors' claims add up to more than is left. To some extent, this can also be applied to taxation issues, where income differentials are similar to claims. But in many problems of fair division, the claims consist in non-transferable differences that affect the benefit the agents can derive from the resource to be shared. In such cases, the claims cannot be directly translated into amounts requested.

For instance, when a group of agents share the cost of a facility, it is commonly the case that differences of distance, access, type, and amount of use justify unequal shares in a way that cannot be dealt with simply by registering monetary claims or by applying a model of cost sharing with equal claims. Consider the following example. An elevator undergoes a costly repair, whose cost has to be shared among the building inhabitants. It is likely that their willingness to pay for the use of the elevator depends on many factors, such as the floor at which they live, their frequency of

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going in and out, their bodily mobility, their income, and their genuine preference regarding stairs vs lift. Would the residents accept the use of a standard cost sharing model? That would imply that, for instance, the disabled tenant of the third floor should pay a big share, while the person living on the sixth floor would pay little just because he greatly values not being bothered by charity demands and environmental activists (who come more often when the elevator is operational). A more reasonable method should instead sort out among these factors those that should not affect the value derived by an agent from the repair (including cost paid): floor, possibly income, and rate of use. The cost shares should then compensate for the influence of these factors on the willingness to pay, so that the persons living on a high floor, or with a high income or frequent use of the lift, should pay more. In contrast, there would be no compensation for the other factors, and it might be deemed normal that the disabled person on the third floor derive a high utility from the repair and that the tenant on the sixth floor derive little utility from it. But this selective approach cannot be easily translated into differential monetary claims. A more comprehensive model seems necessary in order to take account of the relevant information.

In this paper, I will not address the difficult question of how the factors should be sorted out¹ and will examine only the methods by which the partial compensation can be performed. Moreover, I will restrict my attention to the basic case of a pure exchange economy, in which a one-dimensional resource has to be divided among a finite population. Each agent is characterized by a personal parameter which denotes a non-transferable talent or handicap that must be compensated for by an adequate allocation of external resource. The agents have preferences over bundles of external resources *cum* handicaps, so that they can assess how well off they would be with other possible personal parameters and other amounts of external resource.

In a companion paper [4], I introduce this model in more detail and discuss its relevance not only for microeconomic fair division problems, but also for the issue of equality of resources vs equality of welfare (see, e.g., Roemer [9]) and the concept of equal opportunities.² All solutions studied there are inspired by the no-envy approach to equity. Here, three other solutions are examined, which are not directly related to no-envy. One of them, Pazner and Schmeidler's egalitarian-equivalent solution, is well known in the fairness literature. A characterization result is given for each of them.

¹ The classification made in the above example has only an illustrative purpose and may not satisfy the reader.

² A precise application of this approach to the equal opportunity concept is made in Fleurbaey [5].

Here substantial use is made of the axiom of (subgroup) consistency. The no-envy solution is consistent, but turns out to be generally empty in this model (see Fleurbaey [4]). The main solutions which are based on weakened definitions of no-envy and display reasonable existence conditions, however, are not consistent. In this paper, it is shown that the consistency axiom has strong implications in any one-good model and that, nonetheless, reasonable solutions can be characterized in this model if consistency is combined with other mild ethical requisites.

The paper is organized as follows. The next section presents the model and the notation. The main properties that are desirable for a solution in this context are presented in Section 3. Section 4 proceeds with the analysis of the implications of consistency. Section 5 then presents a dual characterization of two consistent solutions. Section 6 introduces and characterizes a third solution, which is not consistent but satisfies other appealing requisites.

2. THE MODEL

Consider a pure exchange economy, in which a given amount ω ($\omega \in \mathbb{R}_{++}$) has to be shared among n agents. Each agent i is characterized by a parameter y_i , which represents her non-transferable talent or handicap, and by preferences R_i (strict preferences P_i and indifference I_i) over pairs (x_i, y_i) in $\mathbb{R}_+ \times Y$. The set Y contains all possible parameters y that the agents may have. No particular mathematical structure is assumed for this set; we only assume that it contains at least three elements. The preference relation R_i is assumed to be strictly monotonic w.r.t. x_i , and representable by a utility function defined over $\mathbb{R}_+ \times Y$, and continuous in x_i . Let $\mathcal{R}$ denote the set of such preferences.³

Let $y = (y_1, \dots, y_n)$, $R = (R_1, \dots, R_n)$ denote the population profiles of handicaps and preferences. An economy is denoted $\mathcal{E} = (y; R; \omega)$ (the number of agents is given by the dimension of y and R). Let $\mathcal{E}$ denote the domain of economies satisfying the above assumptions.

A solution φ is a function defined over a set D_φ of economies, such that for any economy $\mathcal{E}$ of this set, $\varphi(\mathcal{E}) \in \mathbb{R}_+^n$ and $\sum_{i=1}^n \varphi_i(\mathcal{E}) = \omega$. In other words, in this paper only single-valued⁴ solutions yielding feasible allocations are considered.

Let z be an n vector, G a subset of $\{1, \dots, n\}$, and k an n vector of natural numbers. Then $\#G$ denotes the cardinal of G , z_G denotes the $\#G$ vector

³ The ordinal framework adopted here is not essential to the results. See below.

⁴ Single-valuedness is not a strong requirement in a one-good model with monotone preferences.

of components of z which belong to G , and $z^{(k)}$ denotes the $\sum_{i=1}^n k_i$ vector obtained from z by replicating k_i times the i th component of z . It is assumed throughout this paper that the domain D_φ is such that if $\mathcal{E} = (y; R; \omega) \in D_\varphi$, then $(y_G; R_G; \sum_{i \in G} \varphi_i(\mathcal{E})) \in D_\varphi$, $(y^{(k)}; R^{(k)}; \sum_i k_i \varphi_i(\mathcal{E})) \in D_\varphi$, and $(y; R; \omega') \in D_\varphi$, for any $\omega' > \omega$. All solutions considered in this paper have domains satisfying these assumptions.

3. PROPERTIES OF SOLUTIONS

This section introduces the properties of solutions that will appear in subsequent analysis. The first one is a standard anonymity requisite.

Anonymity. If σ is a permutation over $\{1, \dots, n\}$, for any economy $\mathcal{E} = (y; R; \omega)$ in D_φ ,

$$\sigma(\varphi(\mathcal{E})) = \varphi(\sigma(y); \sigma(R); \omega).$$

Another requisite is continuity w.r.t. total resources.

Continuity. The function φ is continuous in ω over D_φ .

The consistency axiom means that the allocation selected in a given economy is still selected if only a subeconomy is considered.

Consistency. For any $G \subset \{1, \dots, n\}$, and any $\mathcal{E} = (y; R; \omega)$ in D_φ ,

$$\varphi_G(\mathcal{E}) = \varphi\left(y_G; R_G; \sum_{i \in G} \varphi_i(\mathcal{E})\right).$$

The axiom of expansion invariance roughly has a converse meaning: any allocation selected by the solution can be retained in a larger economy obtained by addition of agents identical to some agents of the original economy.

Expansion invariance. For any k in $\mathbb{N}^n$, and any $\mathcal{E} = (y; R; \omega)$ in D_φ ,

$$\varphi^{(k)}(\mathcal{E}) = \varphi\left(y^{(k)}; R^{(k)}; \sum_i k_i \varphi_i(\mathcal{E})\right).$$

When $k = (m, \dots, m)$, we get the property of replication invariance (Thomson [15]), stipulating that the replicated allocation of a solution is the solution allocation for the replicated economy.

The next two axioms express an idea of solidarity among the agents of an economy. If the total resources expand, or if the number of agents decreases,

all (remaining) agents should be weakly better off. The former idea has been introduced by Roemer [10], and the latter by Thomson [14].⁵

Resource monotonicity. For any $\mathcal{E} = (y; R; \omega)$ and $\mathcal{E}' = (y; R; \omega')$ in D_φ ,

$$\omega \geq \omega' \Rightarrow \varphi(\mathcal{E}) \geq \varphi(\mathcal{E}').$$

Population monotonicity. For any $\mathcal{E} = (y; R; \omega)$ in D_φ and $G \subset \{1, \dots, n\}$,

$$\varphi_G(\mathcal{E}) \leq \varphi(y_G; R_G; \omega).$$

The properties to be introduced now are more specific to this model. They express the aim of compensating *only* but *fully* for the handicap differentials of the agents.⁶ If the compensation is made only for handicaps, then two agents with identical handicaps should receive the same amount of resource:

Equal resource for equal handicap (EREH). $\forall \mathcal{E} \in D_\varphi, \forall i, j, y_i = y_j \Rightarrow \varphi_i(\mathcal{E}) = \varphi_j(\mathcal{E})$.

On the other hand, if full compensation is performed, then two agents who differ only in their handicaps should obtain the same welfare, or at least, the allocation should maximin their welfare:⁷

Equal welfare for equal preference (EWEP). $\forall \mathcal{E} \in D_\varphi, \forall i, j, R_i = R_j \Rightarrow (\varphi_i(\mathcal{E}), y_i) I_i(\varphi_j(\mathcal{E}), y_j)$, or $\varphi_i(\mathcal{E}) = 0$ and $(0, y_i) R_i(\varphi_j(\mathcal{E}), y_j)$, or $\varphi_j(\mathcal{E}) = 0$ and $(0, y_j) R_j(\varphi_i(\mathcal{E}), y_i)$.

A fundamental limitation to the ethical goal of compensation, in the current framework, is that these two axioms are in general incompatible: no solution defined on a reasonably large domain can compensate strictly and fully for handicap differentials.

PROPOSITION 1. *There is no solution on $\mathcal{D}$ satisfying EREH and EWEP.*

*Proof.*⁸ The counterexamples given in the proof of Fleurbaey [4, Proposition 1] can be adapted easily to the present context. Here is a simple numerical example. Consider $\mathcal{E} = ((1, 1, 3, 3); (R, R', R, R'); 5)$, with R and R' respectively represented by $u(x, y) = x + y$ and $u'(x, y) = x + 2y$. By EREH, $x_1 = x_2$ and $x_3 = x_4$. If EWEP is to be satisfied, there are

⁵ For vectors, the signs $\geq$ and $\leq$ mean that the same holds for all components.

⁶ They are discussed at length in Fleurbaey [4]. See also Iturbe and Nieto [6].

⁷ This weakens the definition of EWEP given in Fleurbaey [4] and Iturbe and Nieto [6]. A further weakening would be possible if comparable utility functions were used: the maximin of welfare might be asked only for two agents displaying the same utility functions. For further reference, let us call this weaker axiom EWEU.

⁸ The proof actually shows that EREH and EWEU are incompatible, which means that adding utility information does not affect this result.

several possibilities. If all x 's are positive, one must have $u(x_1, y_1) = u(x_3, y_3)$, i.e., $x_1 = x_3 + 2$, and $u'(x_1, y_1) = u'(x_3, y_3)$, i.e., $x_1 = x_3 + 4$. That is impossible. If $x_1 = 0$, $u(x_1, y_1) \geq u(x_3, y_3)$ implies $0 \geq x_3 + 2$, which is impossible. If $x_3 = 0$, one must have $u(x_1, y_1) \leq u(x_3, y_3)$, i.e., $x_1 \leq 2$. But then $2x_1 + 2x_3 \leq 4 < \omega = 5$. EWEP cannot be satisfied. ■

As a consequence, it is useful to consider weakened versions of these axioms, which require compensation "only" or "fully", only when all agents have the same handicap or the same preferences:

EREH*. $\forall \mathcal{E} \in D_\varphi, [\forall i, j, y_i = y_j] \Rightarrow [\forall i, j, \varphi_i(\mathcal{E}) = \varphi_j(\mathcal{E})]$.

EWEP*. $\forall \mathcal{E} \in D_\varphi, [\forall i, j, R_i = R_j] \Rightarrow [\forall i, j, (\varphi_i(\mathcal{E}), y_i) I_i(\varphi_j(\mathcal{E}), y_j), \text{ or } \varphi_i(\mathcal{E}) = 0 \text{ and } (0, y_i) R_i(\varphi_j(\mathcal{E}), y_j), \text{ or } \varphi_j(\mathcal{E}) = 0 \text{ and } (0, y_j) R_j(\varphi_i(\mathcal{E}), y_i)]$.

Many reasonable solutions satisfy these two requisites simultaneously.

4. CONSEQUENCES OF CONSISTENCY

The consistency axiom, in a one-good model, has strong consequences in terms of invariance, continuity, monotonicity, and compensation. Propositions 2 through 5 below apply to any one-good model, not only to the current one.

PROPOSITION 2. *Any anonymous consistent solution is expansion invariant.*

Proof. Take any k in $\mathbb{N}^n$ and any economy $\mathcal{E} = (y; R; \omega)$ in D_φ . Let $k^* = \max\{k_1, \dots, k_n\}$ and $\bar{k} = (k^*, \dots, k^*)$. Consider the economy $\mathcal{E}^{(\bar{k})} = (y^{(\bar{k})}; R^{(\bar{k})}; k^*\omega)$. In view of anonymity, necessarily there is an allocation $x \in \mathbb{R}_+^n$ such that $\varphi(\mathcal{E}^{(\bar{k})}) = x^{(\bar{k})}$. Due to consistency, one must have $x = \varphi(\mathcal{E})$. The economy $(y^{(\bar{k})}; R^{(\bar{k})}; \sum_i k_i \varphi_i(\mathcal{E}))$ is then a subeconomy of $\mathcal{E}^{(\bar{k})}$ as defined in the consistency property, so that applying consistency again, one gets a solution allocation for this subeconomy which equals $\varphi^{(\bar{k})}(\mathcal{E})$. ■

Take some given values of y and R . Then the solution φ is said here to be Lipschitzian if each component of the function $\varphi(y; R; \cdot)$ is Lipschitzian on the set $\{\omega \mid (y; R; \omega) \in D_\varphi\}$. Similarly, "the set of discontinuity points of the solution" is a short expression standing for the set of discontinuity points of the function $\varphi(y; R; \cdot)$. An element of $X \subset \mathbb{R}$ is said to be *isolated* if there is a neighborhood of this point containing no other element of X .

PROPOSITION 3. *If a solution is anonymous and consistent, then, for any (y, R) , either it is Lipschitzian or its set of discontinuity points has no isolated point.*

The proof is given in the Appendix. This proposition shows that consistency (with anonymity) precludes all cases of weak continuity or weak discontinuity. Therefore, to combine consistency and continuity in a one-good model is somewhat redundant. It is enough to resort to a mild regularity condition stipulating that the set of discontinuity points of φ , for any (y, R) , is empty or has an isolated point. Let us say that a solution is *regular* if it satisfies this condition.

The next lemma is then a corollary of Proposition 3 and of a result obtained by Young on the bankruptcy model [17, Lemma 2]. A direct proof is given in the Appendix.

PROPOSITION 4. *Any anonymous, consistent, regular solution is resource monotonic.*

It may be asked whether the regularity condition is actually necessary in this proposition, and whether the discontinuous case in Proposition 3 is possible. This is indeed the case, and here is an example of a solution which, for some economies, is neither continuous nor resource monotonic, although it is always anonymous and consistent. Partition the set Y into two subsets Y_1 and Y_2 . For a given profile y , let $\alpha_1(y) = \#\{i \mid y_i \in Y_1\}$ and $\alpha_2(y) = \#\{i \mid y_i \in Y_2\}$. The solution is defined as follows. For any economy $\mathcal{E}$, if $\omega \in \mathbb{Q}$, then $\varphi(\mathcal{E}) = (\omega/n, \dots, \omega/n)$. If $\omega \in \mathbb{R} \setminus \mathbb{Q}$, $\varphi_i(\mathcal{E}) = (\omega + \alpha_1(y) - \alpha_2(y))/n - 1$ if $y_i \in Y_1$, and $\varphi_i(\mathcal{E}) = (\omega + \alpha_1(y) - \alpha_2(y))/n + 1$ if $y_i \in Y_2$.

The representation theorems of Young [17] can be translated into the current framework, but more structure must be assumed for the set Y . This is not examined here, because the representations obtained are still quite general and do not have a direct ethical meaning.

PROPOSITION 5. *Any consistent solution satisfying resource monotonicity is also population monotonic.*

Proof. Consider an economy $\mathcal{E} = (y; R; \omega)$. Suppose p agents are added to the population, with profiles y' and R' . Let $\mathcal{E}' = (y, y'; R, R'; \omega)$. Necessarily $\omega' = \sum_{i=1}^n \varphi_i(\mathcal{E}) \leq \omega$. By consistency, for any $i \leq n$, $\varphi_i(\mathcal{E}') = \varphi_i(y; R; \omega')$. By resource monotonicity, $\varphi_i(\mathcal{E}') \leq \varphi_i(\mathcal{E})$. This entails population monotonicity. ■

This result is in fact true for a large class of economic models (with several goods, production, etc.). It is also easy to check that, in the present model, resource monotonicity alone implies that the solution is Lipschitzian. It is worth noting that, contrary to what happens in most models of fair division, in this model the monotonicity properties are more easily satisfied than no-envy: they are practically implied by consistency, whereas no-envy (which implies EREH and EWEP) is generally empty. The incompatibility between EREH and EWEP is even strengthened by consistency.

PROPOSITION 6. *Any consistent solution satisfying EREH* (resp. EWEP*) satisfies EREH (resp. EWEP).*

Proof. Consider two agents i and j . If $y_i = y_j$, then by EREH* $x_i = x_j$ in the reduced economy formed by these two agents, and by consistency this allocation (x_i, x_j) is the same as that in the full economy.

If $R_i = R_j$, then by EWEP* (x_i, x_j) maximins the welfare of i and j in the subeconomy formed by these two agents, and by consistency again this is also true in the full economy. ■

An important consequence of this proposition is the following corollary of Proposition 1:

COROLLARY 1. *There is no consistent solution on $\mathcal{D}$ satisfying EREH* and EWEP*.*

Therefore if one seeks to compensate as strictly and as fully as possible with a consistent solution, further weakening of the EREH and EWEP requisites is needed. This is the topic of the next section.

5. A DUAL CHARACTERIZATION OF TWO SOLUTIONS

Assume that a benchmark handicap $\tilde{y} \in Y$ and a benchmark preference ordering $\tilde{R} \in \mathcal{R}$ are chosen. One can further weaken EREH* and EWEP* so as to require equal resource or equal welfare only when all agents have the benchmark parameter.

$\tilde{y}$ -EREH*. $\forall \mathcal{E} \in D_\varphi, [\forall i, y_i = \tilde{y}] \Rightarrow [\forall i, j, \varphi_i(\mathcal{E}) = \varphi_j(\mathcal{E})]$.

$\tilde{R}$ -EWEP*. $\forall \mathcal{E} \in D_\varphi, [\forall i, R_i = \tilde{R}] \Rightarrow [\forall i, j, (\varphi_i(\mathcal{E}), y_i) \tilde{I}(\varphi_j(\mathcal{E}), y_j), \text{ or } \varphi_i(\mathcal{E}) = 0 \text{ and } (0, y_i) \tilde{R}(\varphi_j(\mathcal{E}), y_j), \text{ or } \varphi_j(\mathcal{E}) = 0 \text{ and } (0, y_j) \tilde{R}(\varphi_i(\mathcal{E}), y_i)]$.

It may be noted that it would not be of much use to strengthen these axioms so as to require equal resource (or welfare) for two benchmark values of the parameters. For instance, equal resource would be required if all agents had the handicap $\tilde{y}_1$ or if they all had $\tilde{y}_2$. The same counter-example as in the proof of Proposition 1 can in fact be used to show that, with such strengthening, EREH* and $\tilde{y}_{1,2}$ -EWEP*, or $\tilde{R}_{1,2}$ -EREH* and EWEP*, are then incompatible (under consistency).

The benchmark parameters can also serve to define two solutions. The first one, *conditional equality*, has been introduced in a taxation model by Roemer [11]. Although Roemer's formulation is rather complex due to the model he studies, one can derive a much simpler definition in the current framework.

$\tilde{R}$ -Conditional equality. Choose x such that $\forall i, j, (x_i, y_i) \tilde{I}(x_j, y_j)$, or $x_i = 0$ and $(0, y_i) \tilde{R}(x_j, y_j)$, or $x_j = 0$ and $(0, y_j) \tilde{R}(x_i, y_i)$.

This solution simply applies the leximin⁹ criterion referring to $\tilde{R}$, and leximin here coincides with maximin. Indeed, all agents who have a higher welfare (measured by any representation of $\tilde{R}$) than the minimum level have a zero resource and then no redistribution can be made to increase the minimum level of welfare or any intermediate level (without decreasing the minimum level).

This solution may also be interpreted as representing Sen's criterion of equal capabilities.¹⁰ Assume there is a function C which measures the capability index of a person, depending on her resources (x, y) . This function directly defines a preordering like $\tilde{R}$, and equalizing the capability index across agents amounts to enforcing the conditional equality solution. It may look strange that this solution is independent of the preference profile of the population, but this is mainly because there is only one transferable good. In a more general setting, this solution would only define the distribution of *income* independently of the agents' personal preferences about handicaps,¹¹ but of course the allocation of transferable *goods* would depend on the agents' preferences over these goods.

PROPOSITION 7. *The $\tilde{R}$ -conditional equality solution is non-empty over $\mathcal{D}$. It satisfies anonymity, continuity, consistency, expansion invariance, population monotonicity, resource monotonicity, EREH, and $\tilde{R}$ -EWEP*.*

The proof is in the Appendix.

The other solution is the *egalitarian-equivalent* solution proposed by Pazner and Schmeidler [8]. Applied to this context, the common bundle with which every agent is (at least) indifferent is an extended bundle containing the benchmark handicap.

$\tilde{y}$ -Egalitarian-equivalent. Choose x such that $\exists \tilde{x}, \forall i, (x_i, y_i) I_i(\tilde{x}, \tilde{y})$, or $x_i = 0$ and $(0, y_i) R_i(\tilde{x}, \tilde{y})$.

Define the function $\gamma(y_i, R_i) = \inf\{x_i \in \mathbb{R}_+ \mid (x_i, y_i) R_i(0, \tilde{y})\}$. By convention, if this set is empty, then $\gamma(y_i, R_i) = +\infty$. Let $\mathcal{R}'$ be the subset of $\mathcal{R}$ whose elements R_i satisfy

$$\forall x \in \mathbb{R}_+, \forall y \in Y, \exists x' \in \mathbb{R}_+, (x', \tilde{y}) R_i(x, y).$$

This assumption simply excludes preferences for which $\tilde{y}$ is an overwhelming handicap that no amount of external resource can compensate. We can then define the domains $\mathcal{D}' = \{\mathcal{E} \in \mathcal{D} \mid R \in \mathcal{R}'\}$ and $\mathcal{D}'' = \{\mathcal{E} \in \mathcal{D}' \mid \sum_{i=1}^n \gamma(y_i, R_i) \leq \omega\}$.

⁹ On this criterion, see, e.g., Sen [13]. Note that in a framework with utility functions, nothing has to be changed in the definition.

¹⁰ See Sen [12].

¹¹ But not necessarily independently of their opinions: there may be a political process which decides on the function C democratically.

PROPOSITION 8. *The $\tilde{y}$ -egalitarian-equivalent solution is non-empty for $\mathcal{E} \in \mathcal{D}'$ if and only if $\mathcal{E} \in \mathcal{D}''$. It satisfies anonymity, continuity, consistency, expansion invariance, population monotonicity, resource monotonicity, $\tilde{y}$ -EREH*, and EWEP.*

The proof is in the Appendix.

These two solutions have a symmetric behavior as regards the compensation properties (compensation “only” and “fully”). As a matter of fact, and this is the main result of this paper, they can be dually characterized by this particular feature.¹²

PROPOSITION 9. *$\tilde{R}$ -Conditional equality is the only consistent solution over $\mathcal{D}$ which satisfies EREH* and $\tilde{R}$ -EWEP*. The $\tilde{y}$ -egalitarian-equivalent solution is the only consistent solution over $\mathcal{D}''$ which satisfies $\tilde{y}$ -EREH* and EWEP*.*

Proof. Let $\mathcal{E} = (y; R; \omega) \in \mathcal{D}$. Let $\tilde{R}^{(n)}$ denote the n vector $(\tilde{R}, \dots, \tilde{R})$. Consider the economy $\mathcal{E}' = (y, y; R, \tilde{R}^{(n)}; 2\omega)$. Due to Proposition 6, the solution φ satisfies EREH, and therefore there is $x' \in \mathbb{R}_+^n$ such that $\varphi(\mathcal{E}) = (x', x')$. In the reduced economy $(y; \tilde{R}^{(n)}; \omega)$, by $\tilde{R}$ -EWEP*, φ yields an allocation x such that $\forall i, j, (x_i, y_i) \tilde{I}(x_j, y_j)$ or $[x_i = 0 \text{ and } (0, y_i) \tilde{R}(x_j, y_j)]$ or $[x_j = 0 \text{ and } (0, y_j) \tilde{R}(x_i, y_i)]$. Now, by consistency, $x = x' = \varphi(\mathcal{E})$. In conclusion, φ coincides with the conditional equality solution.

Let $\mathcal{E} = (y; R; \omega) \in \mathcal{D}''$. Let $\tilde{y}^{(n)}$ denote the n vector $(\tilde{y}, \dots, \tilde{y})$. By Proposition 8 there exists an egalitarian-equivalent allocation x for $\mathcal{E}$. Let $f(x)$ be defined as in the proof of Proposition 8 and consider the economy $\mathcal{E}' = (y, \tilde{y}^{(n)}; R, R; \omega + nf(x))$. Since $\sum_{i=1}^{2n} \gamma(y_i, R_i) = \sum_{i=1}^n \gamma(y_i, R_i) \leq \omega \leq \omega + nf(x)$, $\mathcal{E}'$ belongs to $\mathcal{D}''$. Let $x' = \varphi(\mathcal{E}')$ and $x'' = (x'_1, \dots, x'_n)$.

By consistency and $\tilde{y}$ -EREH*, for all $i, j > n$, $x'_i = x'_j = \tilde{x}$ for some $\tilde{x}$.

By Proposition 6, φ satisfies EWEP and then, for all $i \leq n$, $(x'_i, y_i) I_i(\tilde{x}, \tilde{y})$ or $[x'_i = 0 \text{ and } (0, y_i) R_i(\tilde{x}, \tilde{y})]$ or $[\tilde{x} = 0 \text{ and } (0, \tilde{y}) R_i(x'_i, y_i)]$.

Consider the possibility of $\tilde{x} = 0$. Then for all $i \leq n$, either $(0, \tilde{y}) R_i(x'_i, y_i)$ or $x'_i = 0$ and $(0, y_i) R_i(0, \tilde{y})$. In both cases, $\gamma(y_i, R_i) \geq x'_i$. We then have $\omega \geq \sum_{i \leq n} \gamma(y_i, R_i) \geq \sum_{i \leq n} x'_i = \omega + nf(x) \geq \omega$. This implies $\gamma(y_i, R_i) = x'_i$ for all i , and $f(x) = 0$. But that means that either $(0, \tilde{y}) I_i(x'_i, y_i)$ or $x'_i = 0$ and $(0, y_i) R_i(0, \tilde{y})$. But this property is precisely satisfied by x , and therefore $x'' = x$. By consistency, φ coincides with the egalitarian-equivalent solution on $\mathcal{E}$.

¹² The characterizations can be extended in a straightforward way to the model with comparable utility functions, using the related versions of EWEU instead of EWEP. Since the two solutions remain ordinal anyway, this means that ordinalism is implied by the axioms, even if it is not assumed in the model.

Now examine the case $\tilde{x} > 0$. We have, for all $i \leq n$, $(x'_i, y_i) I_i(\tilde{x}, \tilde{y})$ or $[x'_i = 0 \text{ and } (0, y_i) R_i(\tilde{x}, \tilde{y})]$. But recall that, for all $i \leq n$, $(x_i, y_i) I_i(f(x), \tilde{y})$ or $[x_i = 0 \text{ and } (0, y_i) R_i(f(x), \tilde{y})]$. Therefore, if $\tilde{x} \geq f(x)$, necessarily $x'_i \geq x_i$ for all i , and then $\sum_{i \leq n} x'_i + n\tilde{x} \geq \omega + nf(x)$. But since $\sum_{i \leq n} x'_i + n\tilde{x} = \omega + nf(x)$, one must have $x'_i = x_i$ for all i . The same reasoning applies in the case $\tilde{x} \leq f(x)$. As a consequence, $x'' = x$. By consistency, φ must coincide with the egalitarian-equivalent solution. ■

In fact, much more than Proposition 9 is proved here. Any solution satisfying the assumptions of the first assertion of the proposition, and non-empty for $\mathcal{E} = (y; R; \omega)$ and $(y, y; R, \tilde{R}^{(n)}; 2\omega)$, must coincide with conditional equality on $\mathcal{E}$. A similar statement holds for the egalitarian-equivalent solution. As a consequence, there is no risk that by restraining the domain assumptions other reasonable solutions satisfying the assumptions of the theorem can be found.

In contrast, it is easy to check that if consistency or one of the compensation properties is dropped, then the result no longer holds and other solutions can be imagined.

6. A THIRD SOLUTION

It is possible to find solutions satisfying EREH and EWEP*, or EREH* and EWEP, simultaneously, but in view of Corollary 1 such solutions cannot be consistent. Solutions of this kind can be derived from conditional equality and the egalitarian-equivalent solution. Indeed, it is enough to make the benchmark parameter ($\tilde{R}$ or $\tilde{y}$) depend on the economy, and more precisely, to choose it among the preferences and handicaps of the agents in the population. The trouble is that if the solution is to be anonymous, the choice of the agent whose parameter will be taken as a benchmark must follow an anonymous procedure, and there may be no obvious candidate in this respect.

This section is devoted to another solution, which satisfies EREH and EWEP*, but is not inspired by conditional equality. A peculiar feature of conditional equality is that it does not take account of the agents' preferences, or, in its revised version just proposed, it uses only one agent's preferences as a "norm." This feature may be useful when the preferences of the agents are not supposed to differ too much from the benchmark and when strong problems of revelation may occur. But in some cases, it may be desirable to register all agents' evaluations of the handicap profile of the population and to aggregate these evaluations in the selection of the allocation of resources. On the other hand, it is also desirable not to rely only on i 's preferences to evaluate y_i , because of the possibility of expensive

tastes and adaptive tastes which might lead i to make unreasonable demands on society's resources, or oppositely to be content with too little. The *average compensation* solution tries to satisfy all these requisites.

Average compensation. Select the allocation

$$x = \frac{1}{n} \sum_{i=1}^n x^i,$$

with $x^i \in \mathbb{R}_+^n$ satisfying $\sum_{j=1}^n x_j^i = \omega$ and, for all j, k , $(x_j^i, y_j) I_i(x_k^i, y_k)$ or $[x_j^i = 0 \text{ and } (0, y_j) R_i(x_k^i, y_k)]$ or $[x_k^i = 0 \text{ and } (0, y_k) R_i(x_j^i, y_j)]$.

In words, each agent i proposes an allocation x^i which leximins the welfare of the population according to her own preferences, and the final allocation that is chosen is simply the mean of such proposals.

PROPOSITION 10. *The average compensation solution is non-empty over $\mathcal{D}$. It satisfies anonymity, continuity, resource monotonicity, EREH, and EWEP*. It does not satisfy consistency, expansion invariance, and population monotonicity.*

The proof of existence is similar to that of Proposition 7. The rest is omitted here.

This solution is sensitive to every agent's valuation of the handicap profile. Here is a brief axiomatic analysis of this issue, and from now on fixed values of y and ω are retained, since we concentrate on responses of the solution to variations of preferences only. To simplify notations, the profile of preferences R will appear as the only argument of a solution $\varphi: \varphi(R)$.

One way to represent valuations of handicaps in view of particular preferences is to use a preordering $\succeq$ over $\mathcal{H} \times \{y_1, \dots, y_n\}$, with $(R_i, y_j) \succeq (R_k, y_l)$, meaning that y_j is deemed a deeper (or equivalent) handicap by R_i than y_l by R_k . A solution φ will be said to be sensitive if it responds to changes of preferences in a way compatible with some preordering of this kind (the preordering may be specific to the solution). That is, if only i 's preferences change, implying a (weak) worsening of her valuation of agent j 's handicap, agent j 's consumption must not decrease, and if there are unanimous preferences in the economy, then a strict worsening of the valuation made by all agents must entail an increase in consumption:

Sensitiveness. There exists a preordering $\succeq$ over $\mathcal{H} \times \{y_1, \dots, y_n\}$, such that for any agents i and j , any profile R , and preferences R'_i , one has

$$(R'_i, y_j) \succeq (R_i, y_j) \Rightarrow \varphi_j(R_{-i}, R'_i) \geq \varphi_j(R)$$

$$(R'_i, y_j) \succ (R_i, y_j) \Rightarrow \varphi_j(R'_i, \dots, R'_i) > \varphi_j(R_i, \dots, R_i).$$

Here is an example of a solution which is sensitive by mimicking a social security system. There is an ex ante allocation $\bar{x}$, which is accepted by the population except that it does not take into account the handicaps y . A group of experts is asked numerical valuations of the various y_j . The weight of agent i in the expert group is α_i , and $\sum_i \alpha_i = 1$. Each expert makes the valuation of handicaps on the basis of her own preferences, and there is a function $f: \mathcal{H} \rightarrow \mathbb{R}_+^n$, with $f_j(R_i)$ being the "points" of handicap given by i to the agent j . A proportional tax is levied on the population, with rate τ , yielding an amount $\tau\omega$ which is distributed to the agents depending on the points allotted by the experts. The final allocation is given by the formula

$$x_j = (1 - \tau) \bar{x}_j + \tau\omega \sum_i \alpha_i \frac{f_j(R_i)}{\sum_k f_k(R_i)}.$$

This solution is sensitive, and the related preordering may be defined by

$$(R_i, y_j) \succeq (R_k, y_l) \Leftrightarrow \frac{f_j(R_i)}{\sum_s f_s(R_i)} \geq \frac{f_l(R_k)}{\sum_s f_s(R_k)}.$$

The sensitiveness axiom is rather weak, since it does not guarantee that the preordering $\succeq$ is well behaved in the sense of giving more resource to an agent if his handicap is really deemed worse (rather than better) by the agents, in terms of welfare. A minimal way to introduce such a guarantee is to require that the solution φ also satisfy EWEP*. Another weakness of the axiom is that it allows great inequalities in the weight of individual preferences, as exemplified by the above solution (in which the expert group may have only one member). An anonymity requirement seems therefore also warranted. But then the following proposition shows that there is not much of a choice as regards the solution.

PROPOSITION 11. *For economies in $\mathcal{E}$ with at least three different handicaps, average compensation is the only anonymous solution satisfying sensitiveness and EWEP*.*

Proof. Let $\hat{\varphi}(R_i)$ denote the leximin function which computes x^i , on the basis of R_i , for the average compensation solution.

Let j , R_i , and R'_i be such that $\hat{\varphi}_j(R'_i) = \hat{\varphi}_j(R_i)$. By EWEP*, $\varphi_j(R'_i, \dots, R'_i) = \varphi_j(R_i, \dots, R_i)$. By the second part of sensitiveness, this implies $(R'_i, y_j) \sim (R_i, y_j)$. By the first part, one deduces $\varphi_j(R_{-i}, R'_i) = \varphi_j(R_{-i}, R_i)$, for any profile R_{-i} of the $n-1$ agents other than i .

This entails that $\varphi_j(R)$ depends only on $\hat{\varphi}_j(R_i)$, $i = 1, \dots, n$. That is, there is a function ψ such that

$$\varphi_j(\mathcal{E}) = \psi_j(\hat{\varphi}_j(R_1), \dots, \hat{\varphi}_j(R_n)).$$

Consider any agent j and any preferences R_j such that $\hat{\phi}_j(R_j) = 0$. By EWEP*, if all agents had preferences R_j , then $\psi(\hat{\phi}_j(R_j), \dots, \hat{\phi}_j(R_j)) = \hat{\phi}_j(R_j)$. But this also reads $\psi_j(0, \dots, 0) = 0$.

By anonymity, ψ must be symmetric. Assume i and j have identical handicaps $y_i = y_j$. By the symmetry of ψ , x_i and x_j must be unchanged if R_i and R_j are permuted. But by anonymity they must be permuted because this is also equivalent to permuting (y_i, R_i) and (y_j, R_j) . Therefore they can only be equal: $x_i = x_j$. In other words, φ satisfies EREH.

Consider the m different handicaps of the population ($m > 2$), and let them be indexed by k . Let h_k be the number of agents endowed with y_k . For any preference R_i , an m vector x^i can be derived from $\hat{\phi}(R_i)$ by letting $x_k^i = h_k \hat{\phi}_i(R_i)$ whenever $y_j = y_k$. One has $\sum_{k=1}^m x_k^i = \omega$. By varying the preferences R_i , x^i can reach any point in the set $\{x \in \mathbb{R}_+^m \mid \sum_{k=1}^m x_k = \omega\}$.

From the function ψ and the EREH property, one can define m functions g_k such that for all k , $g_k(x_k^1, \dots, x_k^n)$ satisfies $\varphi_j(R) = g_k(x_k^1, \dots, x_k^n)/h_k$ whenever $y_j = y_k$. The functions g_k satisfy two properties: $\sum_{k=1}^m g_k(x_k^1, \dots, x_k^n) = \omega$, and $\forall k, g_k(0, \dots, 0) = 0$.

As $m \geq 3$, one can then apply Theorem 2 of Aczél [1, p. 8], and therefore there exist non-negative $\alpha_1, \dots, \alpha_n$, with $\sum_{i=1}^n \alpha_i = 1$, such that for any profile R , $g_k(x_k^1, \dots, x_k^n) = \sum_{i=1}^n \alpha_i x_k^i$. That implies

$$\varphi(R) = \sum_{i=1}^n \alpha_i \hat{\phi}(R_i).$$

By anonymity, one must have $\alpha_i = 1/n$ for all i . ■

It must be emphasized that sensitiveness, and the average compensation solution, do not represent the only way in which subjective assessments of handicaps by the whole population can be aggregated. For instance, a solution proposed by van Parijs [16], and refined by Iturbe and Nieto [6] and Fleurbaey [5], also uses other agents' preferences to assess any given agent's position. Van Parijs' (multi-valued) solution selects allocations such that for any pair of agents i, j , there is at least some agent k such that

$$(x_i, y_i) R_k(x_j, y_j).$$

One sees that this solution is only sensitive to whether, according to agent k 's preferences, (x_i, y_i) is better or worse than (x_j, y_j) . The average compensation solution is in a sense more sensitive to the agents' preferences.

APPENDIX: PROOFS

The following lemma is useful in the proof of Proposition 3.

LEMMA 1. *Any anonymous consistent solution is such that $\forall \mathcal{E} = (y; R; \omega)$, $\mathcal{E}' = (y; R; \omega')$, $\omega \neq \omega'$, $\forall i$,*

$$\frac{\varphi_i(\mathcal{E}) - \varphi_i(\mathcal{E}')}{\omega - \omega'} \in [0, 1] \cup \mathbb{R} \setminus \mathbb{Q}.$$

Proof. Suppose that $[(\varphi_i(\mathcal{E}) - \varphi_i(\mathcal{E}'))/(\omega - \omega')] = p/q$, with p and q integers.

First case: $p > q > 0$. Consider the vector k such that $k_i = p - q$, and for $j \neq i$, $k_j = p$. Expand the economies $\mathcal{E}$ and $\mathcal{E}'$ into $(y^{(k)}; R^{(k)}; p\omega - q\varphi_i(\mathcal{E}))$ and $(y^{(k)}; R^{(k)}; p\omega' - q\varphi_i(\mathcal{E}'))$, respectively. It is easily checked that these two expanded economies are identical. By expansion invariance and Proposition 2, one should have $\varphi_i(\mathcal{E}) = \varphi_i(\mathcal{E}')$, and hence a contradiction.

Second case: $p < 0 < q$. Consider the vector k such that $k_i = -p + q$, and for $j \neq i$, $k_j = -p$. Expand the economies $\mathcal{E}$ and $\mathcal{E}'$ into $(y^{(k)}; R^{(k)}; -p\omega + q\varphi_i(\mathcal{E}))$ and $(y^{(k)}; R^{(k)}; -p\omega' + q\varphi_i(\mathcal{E}'))$, respectively. It is easily checked that these two expanded economies are identical. By expansion invariance and Proposition 2, one should have $\varphi_i(\mathcal{E}) = \varphi_i(\mathcal{E}')$, and hence a contradiction. ■

Proof of Proposition 3. Fix n , y , and R , and let $f = \varphi(y; R; \cdot)$. Suppose that f is continuous on its domain. Choose any i and any x_0 in the domain of f_i . The assumption made in Section 2 entails that $]x_0, +\infty[$ belongs to this domain. By continuity the image set of $]x_0, +\infty[$ by $(f_i(x) - f_i(x_0))/(x - x_0)$ is an interval J . But by Lemma 1, $J \subset [0, 1] \cup \mathbb{R} \setminus \mathbb{Q}$. Therefore, if J is not a subset of $[0, 1]$, J is a singleton $\{\mu\}$, with $\mu > 1$ or $\mu < 0$. Note that if $\mu > 1$ (resp. $\mu < 0$), then for x high enough, $f_i(x) > x$ (resp. $f_i(x) < 0$), which contradicts the feasibility of the solution φ . As a consequence $J \subset [0, 1]$. Since this is true for any x_0 in the domain of f_i , and for any i , that implies that each f_i is Lipschitzian on its domain. Therefore φ is Lipschitzian.

Assume that, on the contrary, f has a discontinuity point x_0 and that f is continuous over $]x_0, x_0 + \varepsilon[$, for some ε . Then, for any i , the function $(f_i(x) - f_i(x_0))/(x - x_0)$ is continuous over $]x_0, x_0 + \varepsilon[$. By Lemma 1, the image of this interval by this function is an interval $J \subset [0, 1] \cup \mathbb{R} \setminus \mathbb{Q}$. But this implies that f is Lipschitzian, and thus that $\lim_{x \rightarrow x_0^+} f_i(x) = f_i(x_0)$. Since this is true for any i , f is right-continuous at x_0 . By similar reasoning, one can show that if f is continuous over $]x_0 - \varepsilon, x_0[$, then f is left-continuous at x_0 . As a consequence, if x_0 is a discontinuity point, then in any interval $]x_0 - \varepsilon, x_0 + \varepsilon[$, there is another discontinuity point. In other words, the set of discontinuity points of f has no isolated point. ■

Proof of Proposition 4. By Proposition 3 the solution is then continuous. In the proof of that proposition, it was in fact shown that in this case, for any pair (x, x') , and any i , $(f_i(x) - f_i(x'))/(x - x') \in [0, 1]$. This directly means resource monotonicity. ■

Proof of Proposition 7. Let the function $\tilde{u}$ represent $\tilde{R}$ over $\mathbb{R}_+ \times Y$ (and be continuous in x). Choose any economy $\mathcal{E}$ in $\mathcal{U}$. For any $x \in \mathbb{R}_+^n$ such that $\sum_{i=1}^n x_i = \omega$, define $f(x) = \min\{\tilde{u}(x_i, y_i) \mid 1 \leq i \leq n\}$. By continuity of $\tilde{u}$, the function f is continuous, over a compact, and therefore attains its maximum in some feasible allocation x^* . Consider an agent i for whom $x_i^* > 0$. If $\tilde{u}(x_i^*, y_i) > f(x^*)$, then a small transfer can be made from i to all agents with the minimum level of welfare $f(x^*)$. Since $\tilde{R}$ is monotonic, with the resulting allocation x one has $f(x) > f(x^*)$, but that is impossible. As a consequence, all agents i such that $x_i^* > 0$ are such that $\tilde{u}(x_i^*, y_i) = f(x^*)$, and all agents i for whom $\tilde{u}(x_i^*, y_i) > f(x^*)$ are such that $x_i^* = 0$. It is immediate to check that this fits the definition of conditional equality. This solution is therefore not empty for $\mathcal{E}$.

The rest of the proof is omitted here. ■

Proof of Proposition 8. Let $\mathcal{E} \in \mathcal{U}''$. The set $K = \{x \in \mathbb{R}_+^n \mid \forall i, x_i \geq \gamma(y_i, R_i), \text{ and } \sum_{i=1}^n x_i = \omega\}$ is then non-empty. It is also compact.

By continuity of the representation of R_i , if $x_i \geq \gamma(y_i, R_i)$, then $(x_i, y_i) R_i(0, \tilde{y})$ and therefore the set $\{\tilde{x}_i \in \mathbb{R}_+ \mid (x_i, y_i) R_i(\tilde{x}_i, \tilde{y})\}$ is not empty. It is also bounded because R_i belongs to $\mathcal{M}'$, and thus it is compact, too, due to the continuity of the representation of R_i . Let $g_i(x_i)$ denote its maximum. It is easy to check that the function g_i is continuous and increasing over the interval $[\gamma(y_i, R_i), \omega]$.

Let $f(x) = \min\{g_i(x_i) \mid 1 \leq i \leq n\}$. This function is well defined and continuous over K . Therefore it attains its maximum at some feasible allocation x^* .

Consider an agent i such that $g_i(x_i^*) > f(x^*)$. If $x_i^* > \gamma(y_i, R_i)$, then a small transfer from i to the agents j such that $g_j(x_j^*) = f(x^*)$ would yield an allocation $x \in K$ such that $f(x) > f(x^*)$. But this is impossible. Now, if $x_i^* = \gamma(y_i, R_i)$ and $x_i^* > 0$, then, by continuity of the representation of R_i , $(x_i^*, y_i) I_i(0, \tilde{y})$, i.e., $g_i(x_i^*) = 0$. This is again impossible. Therefore one must have $x_i^* = 0$.

Therefore, if $x_i^* > 0$, then $g_i(x_i^*) = f(x^*)$, i.e., $(x_i^*, y_i) I_i(f(x^*), \tilde{y})$. And if $g_i(x_i^*) > f(x^*)$, i.e., $(x_i^*, y_i) P_i(f(x^*), \tilde{y})$, then $x_i^* = 0$. In short, x^* satisfies the definition of the egalitarian-equivalent solution.

Conversely, assume that $\sum_i \gamma(y_i, R_i) > \omega$. Assume there is an allocation x and a number $\tilde{x}$ satisfying the definition of the egalitarian-equivalent solution. Necessarily, for all i , $(x_i, y_i) R_i(\tilde{x}, \tilde{y}) R_i(0, \tilde{y})$, which implies $x_i \geq \gamma(y_i, R_i)$. But then x cannot be feasible.

The rest of the proof is omitted here. ■

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